

About IIIT Kottayam

Indian Institute of Information Technology Kottayam at Valavoor, Pala, Kerala is one among the IIITs that have been established as "Institution of National Importance" by MoE, Govt. of India under the ambit of IIIT PPP act 2017. IIIT Kottayam campus is situated at Valavoor, Pala, Kottayam - a picturesque location, spread over 53 acres of land - allotted by Govt. of Kerala in the southern part of the Country commonly referred to as God's Own Country. B.Tech students are admitted against the JoSAA/CSAB counselling based on the JEE main rank list. The Institute offers 4 year B.Tech and B.Tech (Hons) programs in ECE, CSE, AI and Data Science, and Cybersecurity. In addition, the Institute offers Ph.D. programs in ECE, CSE and Mathematics as well as M.Tech programme in AI & Data Science, Cyber Security, CSE with specialization in Big Data and ML.

About Network Science Research Lab

Network science explores complex systems represented as networks or graphs, aiding in understanding interactions and behaviors across various domains like sociology, biology, and computer science. It helps uncover patterns and predict behaviors in interconnected systems, playing a crucial role in our increasingly interconnected world. The Network Science Research Group at IIIT Kottayam brings together expertise from Computer Science and Mathematics to create real-world applications, bridging the gap between theory and practice. Our group currently focuses on the robustness of complex networks, navigation in complex networks, representation learning, dynamics of online social networks, bibliographic networks, and functional brain networks.

Visit us at: <https://netsci.iiitkottayam.ac.in/>

About the Workshop

Workshop on Complex Network Analysis with Applications in Brain Network Science and Complex Systems aims to bring together academicians, researchers, industrial experts, Ph.D. scholars, and postdoctoral fellows to explore recent advancements and foundational concepts in the fields of graph theory and its applications in network analysis. Graph theory, a cornerstone of discrete mathematics, offers a robust framework for modeling and analyzing complex networks across various domains from biological systems and brain connectivity to social, technological, and infrastructural networks. The primary aim of this five-day workshop is to provide a comprehensive introduction to the mathematical foundations and computational techniques in complex network analysis, with a particular emphasis on its applications in brain network science and biomedical data analysis. The program will cover a range of contemporary topics, including the simplicial analysis of fMRI data to study human brain dynamics during functional cognitive tasks, analysis of complex networks and prediction using deep learning models, and exploration of graph algorithms along with their computational complexity. Participants will also gain exposure to advanced methodologies such as the application of complex networks in machine learning, the characterization of resting-state fMRI for brain connectivity analysis, and diffusion MRI analysis for clinical applications. In addition, the workshop will introduce recurrence network analysis, which is used to predict climate changes and other dynamic systems. To complement these theoretical discussions, hands-on sessions will be conducted on complex network analysis using NetworkX, and nonlinear dynamics in recurrence relations.

Target Audience

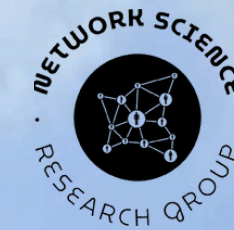
The workshop is intended to equip researchers, students, and practitioners with both foundational understanding and practical skills to address challenges in the analysis of large, complex networks and neuroimaging data

Special Opportunities

- **CSS Members Get FULL Fee Waiver! (Supported by the Complex Systems Society, CSS)**
- **The first 30 student registrations for physical participation will be refunded . (Supported by SERB Matrics Initiative)**
- **Register early to secure your spot and enjoy this exclusive opportunity !**

CNA Brain Workshop Website Link:

<https://sites.google.com/iiitkottayam.ac.in/cna-brain-2025/home>



CNA-Brain-2025 (Hybrid Mode)

Five Day Workshop on

Complex Network Analysis with Applications in Brain Network Science & Complex Systems

19 - 23 December, 2025

Organized by

**Network Science Research Lab
Department of Computer Science and Engineering
Indian Institute of Information Technology
Kottayam, India**

Sponsored by



Keynote Speakers

- **Dr. Neelima Gupte**
Professor
IIT Madras
- **Dr. Somasundaram K**
Professor
Amrita Vishwa Vidhyapeetham,
Coimbatore
- **Dr. Dhrubajyoti Biswas**
Research Scientist-I
NBRC Gurgaon
- **Prof. Neelam Sinha**
Associate Professor
IISc Bangalore
- **Dr. Rohini P**
Assistant Professor
IIITDM Kancheepuram
- **Dr. Jasleen Gund**
Assistant Professor
Amity institute Noida, Uttar Pradesh

Objectives of the Workshop

- To provide a clear and comprehensive understanding of complex network analysis and its diverse applications across scientific domains.
- To strengthen participants' foundational knowledge in graph modeling, graph algorithms, and computational complexity, especially for analyzing large-scale networks.
- To highlight graph theory as a fundamental framework for studying and interpreting the behavior of complex systems.
- To offer hands-on training in designing and analyzing graph-based algorithms using tools such as NetworkX, dynamic graph modeling techniques, and recurrence network analysis.
- To explore the application of complex network analysis in neuroscience, with a focus on understanding brain connectivity using neuroimaging data, while fostering interdisciplinary collaboration among computer science, neuroscience, physics, and data science fields.
- To demonstrate how AI and machine learning approaches can enhance brain network modeling, support clinical prediction, and assist in biomarker identification.



Address

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Valavoor P.O,
Pala, Kottayam-686635, Kerala.
Visit us at: www.iiitkottayam.ac.in

Internal Speakers

- **Dr. Ebin Deni Raj**
Assistant Professor, IIIT Kottayam
- **Dr. Divya Sindhu Lekha**
Assistant Professor, IIIT Kottayam
- **Dr. Christina Terese Joseph**
Assistant Professor, IIIT Kottayam
- **Dr. Nayana P G**
Assistant Professor, IIIT Kottayam
- **Dr. Jisha Mariyam John**
Assistant Professor, IIIT Kottayam
- **Ms. Julie Sara Benny**
Visiting scholar, Purdue University, USA
IIIT Kottayam
- **Dr. Vidyasagar Thalapala**
IIIT Kottayam

Patrons

Prof. Prasad Krishna, Director, IIIT Kottayam
Dr. M Radhakrishnan, Registrar, IIIT Kottayam
Prof. Ashok S, Professor In-charge, Academics, IIIT Kottayam

Organizing Committee

Dr. Divya Sindhu Lekha, Assistant Professor, IIIT Kottayam
Dr. Dhanyamol M V, Assistant Professor, IIIT Kottayam
Dr. Anandhu Mohan, Assistant Professor, IIIT Kottayam
Dr. Nayana P G, Assistant Professor, IIIT Kottayam
Dr. Jisha Mariyam John, Assistant Professor, IIIT Kottayam

Registration for CNA Brain Workshop

<https://forms.gle/vC1rXCwQrXi6oGz67>

Early Bird: 15-12-2025

E-Payment CNA Brain Workshop

<https://www.onlinesbi.sbi/sbicollect/>



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